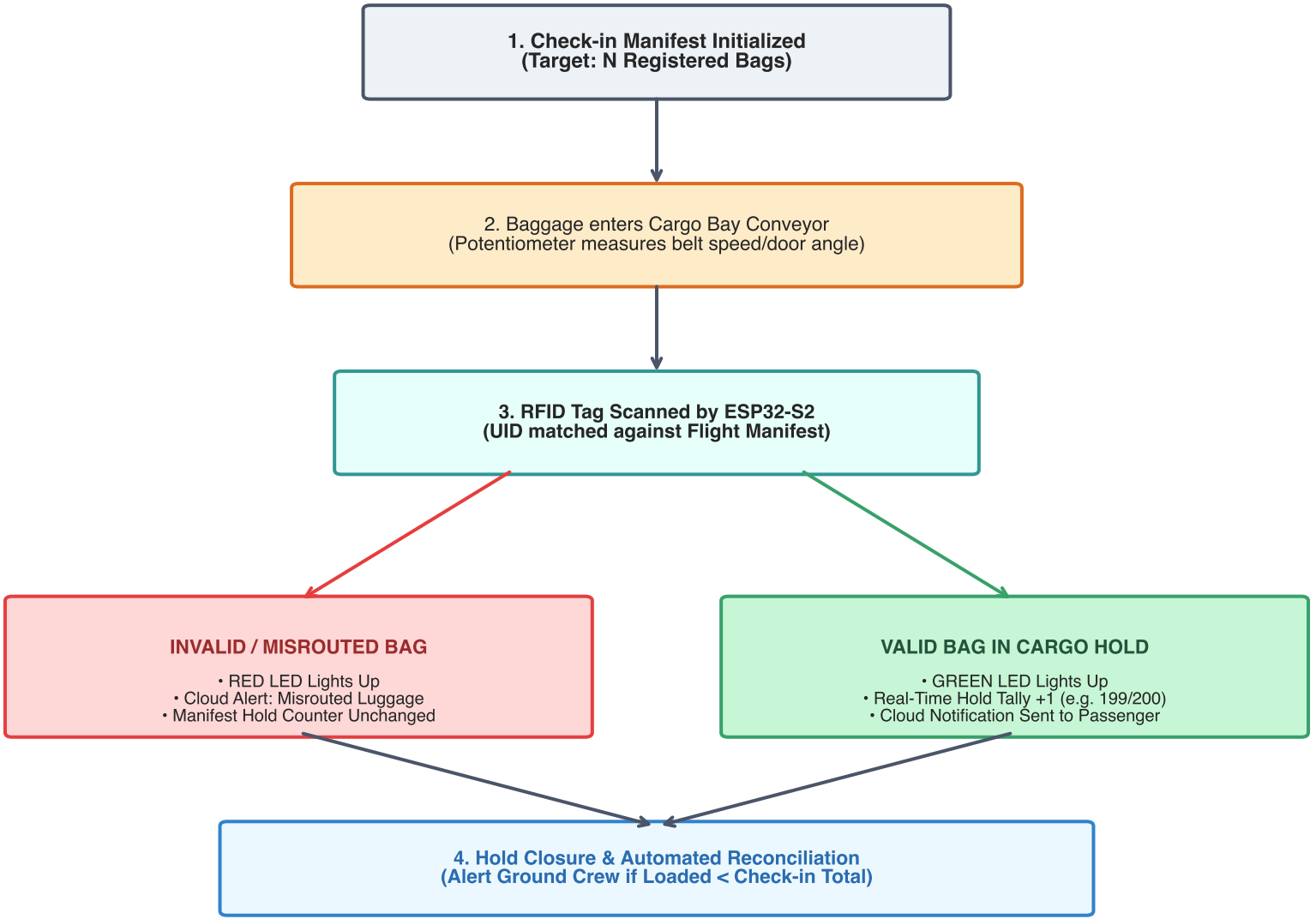
1. Check-in Manifest Initialized
(Target: N Registered Bags)
2. Baggage enters Cargo Bay Conveyor
(Potentiometer measures belt speed/door angle)
3. RFID Tag Scanned by ESP32-S2
(UID matched against Flight Manifest)
INVALID / MISROUTED BAG
• RED LED Lights Up
• Cloud Alert: Misrouted Luggage
• Manifest Hold Counter Unchanged
VALID BAG IN CARGO HOLD
• GREEN LED Lights Up
• Real-Time Hold Tally +1 (e.g. 199/200)
• Cloud Notification Sent to Passenger
4. Hold Closure & Automated Reconciliation
(Alert Ground Crew if Loaded < Check-in Total)